

Week 4: Professional Research skills

How to read this document?

If you have 5 minutes spare for reading, the version of Chapter 4 will give you short, introductory answers to the question:

- Why do research skills matter?
- Why might I conduct research?
- What techniques will improve my research quality?

Full- details can be referred to the FIT1055 textbook.

Only got 5 minutes?

Why do research skills matter?

Research is an essential skill in the modern tech workforce.

Research skills enable you to find information to answer questions, build evidence and solve problems.

Having found information, research skills help you to assess the quality of the information you have found: Is the information relevant? Is it credible? Can you trust it?

Lastly, research skills help you to demonstrate to others how your thinking is supported by good evidence. People are more likely to be persuaded by your ideas if you back them up with quality, relevant, verifiable information.

Employers want people who:

- know how to find useful information (beyond the first page of Google results!)
- can critically evaluate the information they find
- can demonstrate and share their evidence-base with colleagues, clients, and stakeholders

For these reasons and more, research is a crucial area of professional skill development.

Why might I conduct research?

Research can be defined simply as actively finding information new to the researcher or researchers. You are conducting research whenever you have a question and start looking for information that may provide answers.

As an IT professional, you may need to conduct research for various purposes. E.g.:

- Maybe the code you wrote isn't compiling correctly, and you start Googling programming forums looking for a solution.
- Perhaps you need to persuade your boss to change direction on a project, so you look for data to support your argument.
- Maybe you want to understand an emerging technology trend, so you try to find insightful news articles or research papers on the topic

The purpose of your research will determine the research process you take. Try to reflect on the purpose before you start finding and interpreting information.

Compared to research conducted in professional contexts, academic research often involves a more rigorous and specialized process of gathering information and producing new knowledge. Be mindful of this process when conducting your research in an academic setting (i.e., at a university).

What methods can help me develop research skills?

Like any skill, effective and quality research requires technique. In this unit, we will learn the ***intelligent research cycle*** to develop our research skills. The cycle involves the below processes.

1. Collection and gathering of data

Learning the search operators will improve your search results whenever you search for information in an online system. Search operators give you greater control over your search. They help maximize the number of useful search results and filter out less relevant results.

Every search engine or database will have its search operators. Many systems use a variant of Boolean search operators. If searching Google, use the [Google search operators](#). If searching in Monash University Library databases, [learn effective search strategies](#).

2. Processing of data and information

When researching, try to store the information you find in a consistent, logical system. Aim for a logically structured folder system with consistent file naming conventions. Keep track of important information about the sources you have collected, ideally in full APA citations (details to follow). This will make it easier to retrieve relevant information when you need it at a later stage of the research process.

3. Analysis

Analysis and production of the output from processing will include *integrating, evaluating, and analyzing* all available data and information (details to follow). This is a very critical stage that you can follow below to develop your analytical skills at this stage.

Evaluating information sources

Information only has value if you can critically evaluate its credibility. If you trust all the information you find at 'face value, you might build your analysis on unreliable evidence.

You can assess the credibility of information by considering its attributes. The key attributes of information are shown in Table 1 below.

Table 1: Key attributes for evaluation of sources

Currency	How up-to-date is the information? Is the information still relevant and recent for the topic?
Accuracy	Is the information correct? Can you assess how accurate the information is by examining other sources of information as well?
Authority	Who produced the information (author/s or organization), and do they have expertise in the topic? Does the information source provide credible evidence to support its claims?
Accessibility	How easy is it to find this information source or understand its content?(e.g., is the information in a foreign or technical language hidden behind a paywall?)
Stability	How credible is this information to remain the same over time? Is the information or website volatile (e.g., updated or altered frequently) or stable?

Integrating information sources

Writing skills help you integrate the information you have found into your ideas. Two key techniques for integrating sources into your writing are:

- Quotation: using the exact words of another author or source within your text in quotation marks
- Paraphrasing: rephrasing words expressed by another author or source to convey the idea in your own words

Paraphrasing is a particularly helpful technique. It enables you to demonstrate your knowledge of a source, integrate its ideas into the flow of your argument, or convey the information more concisely or understandably.

Whether quoting or paraphrasing another information source, you must acknowledge where the ideas came from with a citation.

Acknowledging information sources

Ethical research practice requires always acknowledging when you have used information from another author or source. At Monash University, we refer to the ethics of academic conduct as academic integrity.

Failing to properly acknowledge your use of information or words produced by others is known as plagiarism – which can carry heavy academic and professional penalties.

In an academic context, citing and referencing is the primary technique for acknowledging your use of information sources. A relevant citation should support every claim or assertion you make in an assignment. Every information source you used in your assignment should be listed in a reference list at the end of the document.

A range of resources can help you develop skills in academic integrity and citing and referencing:

[Monash IT Academic Integrity Moodle site](#)

[Monash IT Style Guide on referencing](#)

[Ask library staff for advice](#) on your citing and referencing

In this unit, we will use the [APA](#) reference citation style.

Dissemination

Dissemination is the process by which the finished product (deliverable) of the work or project is passed to supervisors, bosses, or organizations.

Feedback

Feedback can be a further phase in the intelligence cycle where either the supervisors, boss, or the decision-maker from the initial stage provides feedback to the team, where revised requirements of the process are needed, and a different cycle in the intelligence process is instigated.